

If Statement

```
In [3]: if True:
        print('hello')
```

hello

```
In [4]: if True:
        print('hello')
```

```
Cell In[4], line 2
    print('hello')
    ^
IndentationError: expected an indented block after 'if' statement on line 1
```

```
In [5]: if True:
        print('hello')
```

hello

```
In [6]: if False:
        print('bye')
```

```
In [7]: if True:
        print('Data Science')

        print('bye for now')
```

Data Science
bye for now

```
In [8]: if False:
        print('Data Science')
        print('bye for now')
```

bye for now

```
In [9]: if True:
        print('data science')

        print('bye for now')
```

data science
bye for now

If-Else Statement

```
In [10]: if True:
          print('data science')

          else:
            print('bye for now')
```

data science

```
In [11]: if False:
          print('Data Science')

          else:
          print('bye for now')
```

bye for now

Write Python Code to Check Wheater Number is Even or Odd

```
In [12]: x = 4

          r = x % 2

          if r == 0:
              print('Even Number')
```

Even Number

```
In [13]: x = 5

          r = x % 2

          if r == 0:
              print('even number')
```

```
In [14]: x = 6

          r = x % 2

          if r == 0:
              print('even number')

          if r == 1:
              print('odd number')
```

even number

```
In [15]: x = 6

          r = x % 2

          if r == 0:
              print('Even number')
          else:
              print('odd number')
```

Even number

```
In [16]: x = 6
          r = x % 2
```

```
if r == 0:
    print('even number')

print('odd number')
```

even number
odd number

```
In [17]: x = 4
         r = x % 2

         if r == 0:
             print('even number')
         else:
             print('odd number')
```

even number

```
In [18]: x = 5
         r = x % 2

         if r == 0:
             print('even number')
         else:
             print('odd number')
```

odd number

```
In [19]: x = 10
         r = x % 2

         if r == 0:
             print('even number')
         if r == 1:
             print('odd number')
```

even number

```
In [20]: x = 9
         r = x % 2

         if r == 0:
             print('Even number')

         if r != 0:
             print('odd number')
```

odd number

Nested If Statement

```
In [21]: x = 3
         r = x % 2

         if r == 0:
             print('even number')
```

```
    if x > 5:
        print('greater number')
    else:
        print('odd number')
```

odd number

```
In [22]: x = 6
        r = x % 2

        if r == 0:
            print('Even number')

            if x>5:
                print('greater number')
            else:
                print('smaller number')

        else:
            print('Odd Number')
```

Even number

greater number

```
In [23]: x = 4

        if x == 1:
            print('one')
        if x == 2:
            print('two')
        if x == 3:
            print('three')
        if x == 4:
            print('four')
```

four

If-Elif-Elif Statement

```
In [24]: x = 2

        if x == 1:
            print('one')
        elif x == 2:
            print('Two')
        elif x == 3:
            print('Three')
        elif x == 4:
            print('four')
```

Two

```
In [25]: x = 2

        if x == 1:
            print('one')
```

```
elif x == 2:
    print('Two')
elif x == 3:
    print('Three')
elif x == 4:
    print('four')
```

Two

```
In [26]: x = 10

if x == 1:
    print('one')

elif x == 2:
    print('Two')
elif x == 3:
    print('Three')
elif x == 4:
    print('four')

else:
    print('number not found')
```

number not found

```
In [ ]: num = int(input("Enter a number: "))

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

Geeks for geeks

```
In [28]: age = 20
if age >= 18:
    print("Eligible to vote.")
```

Eligible to vote.

```
In [29]: age = 20
if age >= 18:
    print("Eligible to vote")
```

Eligible to vote

```
In [30]: age = 10
if age <= 12:
    print("Travel for free")
else:
    print("pay for ticket")
```

Travel for free

```
In [31]: marks = 35
res = "Pass" if marks >= 40 else "Fail"
print(f"Result: {res}")
```

Result: Fail

```
In [32]: age = 25

if age <= 12:
    print("Child.")
elif age <= 19:
    print("teenager")
elif age <= 35:
    print("young adult.")
else:
    print("Adult")
```

young adult.

```
In [33]: age = 70
is_member = True

if age >= 60:
    if is_member:
        print("30% senior discount!")
    else:
        print("20% senior discount.")
else:
    print("not eligible for a senior discount.")
```

30% senior discount!

```
In [34]: # Assign a value based on a condition
age = 20
s = "Adult" if age >= 18 else "Minor"
print(s)
```

Adult

```
In [35]: number = 2
match number:
    case 1:
        print("one")
    case 2 | 3:
        print("two or three")
    case _:
        print("other number")
```

two or three

```
In [36]: # Python if Else Statement - Conditional Statement
```

```
In [37]: i = 10
```

```
In [38]: # Checking if i is greater than 15
if (i > 15):
```

```
print("10 is less than 15")

print("i am not in if")
```

i am not in if

```
In [39]: i = 20

# Checking if i is greater than 0
if (i > 0):
    print("i is positive")
else:
    print("i is 0 or negative")
```

i is positive

```
In [40]: a = -2

# Ternary conditional to check if number is positive or negative
res = "Positive" if a >= 0 else "Negative"
print(res)
```

Negative

```
In [41]: age = 25
        exp = 10

# Using '>' operator & 'and' with if-else
if age > 23 and exp > 8:
    print("Eligible")
else:
    print("NOT eligible")
```

Eligible

```
In [42]: i = 10
        if (i == 10):

            # First if statement
            if (i < 15):
                print("i is smaller than 15")

            # Nested - if statement
            # Will only be executed if statement above
            # it is true

            if (i < 12):
                print("i is smaller than 12 too")
            else:
                print("i is greater than 15")
```

i is smaller than 15

i is smaller than 12 too

```
In [43]: i = 25

# Checking if i is equal to 10
if (i == 10):
```

```

    print("i is 10")
    # Checking if i is equal to 15

elif (i == 15):
    print("i is 15")
    # Checking if i is equal to 20

elif (i == 20):
    print("i is 20")

    # If none of the above conditions are true
else:
    print("i is not present")

```

i is not present

In [44]: # if , if..else, Nested if, if-elif statements

In [45]: # if statement example

```

if 10 > 5:
    print("10 greater than 5")

print("Program ended")

```

10 greater than 5
Program ended

In [46]: # if..else statement example

```

x = 3
if x == 4:
    print("Yes")
else:
    print("No")

```

No

In [47]: # if..else chain statement

```

letter = "A"

if letter == "B":
    print("letter is B")

else:

    if letter == "C":
        print("letter is C")

    else:

        if letter == "A":
            print("letter is A")

        else:
            print("letter isn't A, B and C")

```

letter is A

In [48]: *# Nested if statement example*

```
a = 10

if a > 5:
    print("Bigger than 5")

    if a <= 15:
        print("Between 5 and 15")
```

Bigger than 5

Between 5 and 15

In [49]: *# if-elif statement example*

```
letter = "A"

if letter == "B":
    print("letter is B")

elif letter == "C":
    print("letter is C")

elif letter == "A":
    print("letter is A")

else:
    print("letter isn't A, B or C")
```

letter is A

In [50]:

```
x = 10
y = 5
if x > 5:
    if y > 5:
        print("x is greater than 5")
    elif y==5:
        print("x is greater than 5 and y is 5")
    else:
        print("x is greater than 5 and y is less than 5")
```

x is greater than 5 and y is 5

In [52]: *# Nested-if statement*

In [53]:

```
age = 30
member = True

if age > 18:
    if member:
        print("Ticket price is $12.")
    else:
        print("Ticket price is $20.")
else:
    if member:
        print("Ticket price is $8.")
    else:
        print("Ticket price is $10.")
```

Ticket price is \$12.

```
In [54]: i = 0;

# if condition 1
if i != 0:
    # condition 1
    if i > 0:
        print("Positive")

    # condition 2
    if i < 0:
        print("Negative")
else:
    print("Zero")
```

Zero

```
In [55]: # Python If Else in One Line
```

```
In [56]: x = 10
y = 5

res = "x is greater" if x > y else "y is greater"
print(res)
```

x is greater

```
In [57]: n = -5

res = "Positive" if n > 0 else "Negative" if n < 0 else "Zero"

print(res)
```

Negative

```
In [58]: x = 15

res = "Greater than 20" if x > 20 else "Greater than 10" if x > 10 else "10 or less"
print(res)
```

Greater than 10

```
In [59]: # Check multiple conditions in if statement
```

```
In [60]: # Syntax:

#if (condition):
#code1
#else:
#code2
```

```
In [61]: age = 18

if ((age >= 8) and (age <= 12)):
    print("YOU ARE ALLOWED. WELCOME !")
```

```
else:
    print("SORRY ! YOU ARE NOT ALLOWED. BYE !")
```

SORRY ! YOU ARE NOT ALLOWED. BYE !

```
In [62]: var = 'N'

if (var == 'Y' or var == 'y'):
    print("YOU SAID YES")
elif(var == 'N' or var == 'n'):
    print("YOU SAID NO")
else:
    print("INVALID INPUT")
```

YOU SAID NO

```
In [63]: a = 7
b = 9
c = 3

if((a>b and a>c) and (a != b and a != c)):
    print(a, " is the largest")
elif((b>a and b>c) and (b != a and b != c)):
    print(b, " is the largest")
elif((c>a and c>b) and (c != a and c != b)):
    print(c, " is the largest")
else:
    print("entered numbers are equal")
```

9 is the largest

```
In [64]: a = 1
b = 1
c = 1
if(a == 1 and b == 1 and c == 1):
    print("working")
else:
    print("stopped")
```

working

```
In [65]: # Python if AND
```

```
In [67]: a = 20 # age
b = True # Citizen status
if a >= 18 and b:
    print("Eligible")
else:
    print("Ineligible")
```

Eligible

```
In [68]: a = 23 # age
b = "yes" # permission

if a >= 18 and b == "yes":
    print("Granted")
```

```
else:
    print("Denied")
```

Granted

```
In [69]: p = "securePass123" #password
if len(p) >= 8 and any(char.isdigit() for char in p):
    print("Valid")
else:
    print("Invalid")
```

Valid

```
In [70]: a = 50 # health
b = True # has_weapon
if a > 0 and b:
    print("Fight")
else:
    print("No Fight")
```

Fight

```
In [71]: # Python Match Case Statement
```

```
In [72]: def check_number(x):
    match x:
        case 10:
            print("It's 10")
        case 20:
            print("It's 20")
        case _:
            print("It's neither 10 nor 20")

check_number(10)
check_number(30)
```

It's 10

It's neither 10 nor 20

```
In [73]: def greet(person):
    match person:
        case "A":
            print("Hello, A!")
        case "B":
            print("Hello, B!")
        case _:
            print("Hello, stranger!")

greet("A")
greet("B")
```

Hello, A!

Hello, B!

```
In [74]: def num_check(x):
    match x:
        case 10 | 20 | 30: # Matches 10, 20, or 30
            print(f"Matched: {x}")
```

```

        case _:
            print("No match found")

num_check(10)
num_check(20)
num_check(25)

```

Matched: 10
Matched: 20
No match found

```

In [75]: def num_check(x):
        match x:
            case 10 if x % 2 == 0: # Match 10 only if it's even
                print("Matched 10 and it's even!")
            case 10:
                print("Matched 10, but it's not even.")
            case _:
                print("No match found")

num_check(10)
num_check(15)

```

Matched 10 and it's even!
No match found

```

In [76]: def process(data):
        match data:
            case [x, y]:
                # A list with two elements
                print(f"Two-element list: {x}, {y}")
            case [x, y, z]:
                # A list with three elements
                print(f"Three-element list: {x}, {y}, {z}")
            case _:
                print("Unknown data format")

process([1, 2])
process([1, 2, 3])
process([1, 2, 3, 4])

```

Two-element list: 1, 2
Three-element list: 1, 2, 3
Unknown data format

```

In [77]: def person(person):
        match person:

            # Dictionary with name and age keys
            case {"name": name, "age": age}:
                print(f"Name: {name}, Age: {age}")

            # Dictionary with only name key
            case {"name": name}:
                print(f"Name: {name}")
            case _:
                print("Unknown format")

```

```
person({"name": "Alice", "age": 25})
person({"name": "Bob"})
person({"city": "New York"})
```

Name: Alice, Age: 25

Name: Bob

Unknown format

In [78]: *# Ternary Operator in Python*

```
In [79]: n = 5
res = "Even" if n % 2 == 0 else "Odd"
print(res)
```

Odd

```
In [80]: n = -5

res = "Positive" if n > 0 else "Negative" if n < 0 else "Zero"
print(res)
```

Negative

```
In [81]: n = 7
res = ("Odd", "Even")[n % 2 == 0]
print(res)
```

Odd

```
In [82]: a = 10
b = 20
max = {True: a, False: b}[a > b]
print(max)
```

20

```
In [83]: a = 10
b = 20
max = (lambda x, y: x if x > y else y)(a, b)
print(max)
```

20

```
In [84]: a = 10
b = 20

print("a is greater" if a > b else "b is greater")
```

b is greater